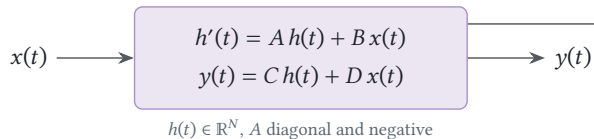
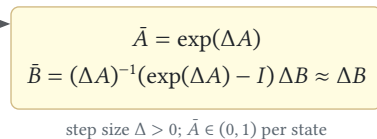


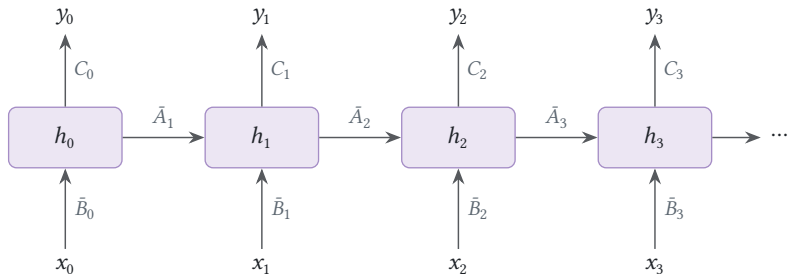
(A) CONTINUOUS STATE SPACE



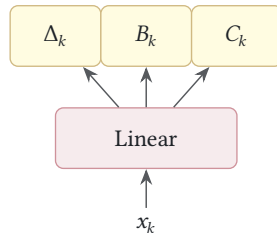
(B) DISCRETISATION, ZERO-ORDER HOLD



(C) RECURRENT VIEW, ONE STEP PER TOKEN



(D) SELECTION (MAMBA)



$\Delta_k \rightarrow 0$: $\bar{A}_k \rightarrow 1$, $\bar{B}_k \rightarrow 0$, the state is held and x_k ignored.

Δ_k large: $\bar{A}_k \rightarrow 0$, the past is forgotten and the state rewritten from x_k .

Because Δ, B, C depend on the input, the model chooses per token what to keep. This is the knob CFFM-Net scales by sensor reliability.

$h_k = \bar{A}_k h_{k-1} + \bar{B}_k x_k$, $y_k = C_k h_k + D x_k$. One pass over L tokens costs $O(L)$, and the state h is all the memory the model keeps.